

Hithaishi Surendra

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Google Scholar

Education

Arizona State University - Tempe, AZ, USA

Aug 2024 – May 2026

Master of Science in Computer Science - GPA: 3.92/4.0

◦ **Coursework:** Machine Learning, Natural Language Processing, Agentic AI, Distributed Systems, Cloud Computing, Data Mining

R.V. College of Engineering - Bangalore, KA, India

Aug 2019 – June 2023

Bachelor of Engineering in Electronics and Telecommunication - GPA: 3.54/4.0

Experience

Software Developer Intern

June 2025 – Jan 2026

Thropic

Denver, CO, USA

- Built **automated scoring backend** integrating **ESPN APIs** (NFL, NBA, NCAA), eliminating manual validation.
- Integrated **Claude** into quiz-generation workflows, designing **prompts and automation pipelines** that **eliminated manual** content creation steps across live campaigns.
- Designed and optimized **Postgres schemas** for players, matches, and scoring, reducing **query latency by 40%**.
- Replaced **Feathery** with internal dashboard built from scratch, enabling self-serve game launches.
- Led **0 to 1** website (**React, Next.js, Tailwind**), scaling to **10K+ users**, **3× engagement**, **2× session time**.

Software Developer Intern

June 2025 – Aug 2025

Vantion

Phoenix, AZ, USA

- Split a **monolithic application** into separate **FastAPI (backend)** and **Next.js (frontend)** services, fixing more than **12 integration issues** and reducing deployment time by **90%**.
- Set up **CI/CD pipelines** on Vercel with automated previews and **sub-minute builds**; used **Claude Code, Cursor, and Copilot** to accelerate development.
- **Updated data handling and APIs** after Supabase schema redesign, improving auth and cutting dev time by 40%.

Associate Software Engineer

July 2023 – July 2024

Telstra

Bangalore, KA, India

- Built **backend services** for a **Kafka-based messaging platform** using **Node.js** and **MongoDB**, supporting over **10K messages per day** with stable uptime.
- Improved reliability and scalability by using **Kafka queues** and **containerized deployments with Docker**.
- **Automated 850+ cloud backups** with **Python** across NetApp and Commvault, improving success rate by **15%**.
- Provisioned **NetApp Storage Virtual Machines** and applied **QoS policies** to maintain steady **IOPS**.

Publications

Lane and Traffic Sign Detection for ADAS using Deep Learning ([Read](#))

IJRITCC, 2023

- Implemented **SegNet** for semantic segmentation (lane detection, **94.80% accuracy, 93.33% mAP**) and **YOLO** for traffic sign detection (**95.56% accuracy**) in the **CARLA** autonomous driving simulator.
- Created custom datasets of **864** (lane) and **612** (traffic sign) images across multiple towns and weather conditions; outperformed **Faster R-CNN** and **R-FCN** baselines.

Deep Learning-based Campus Assistive Chatbot ([Read](#))

IEEE CSITSS, 2021

- Built an **NLP pipeline** using **NLTK, Bag of Words**, and a **Sequential Neural Network** (ReLU + Softmax) for intent classification, achieving **89% accuracy** across 100 test interactions.
- Integrated **Google Maps API** and a feedback loop for iterative accuracy improvement; deployed on college website handling **100+ daily queries** as part of **Cisco internship** at R.V. College of Engineering.

Projects

Agentic AI Systems on TAU-Bench (ASU)

Jan 2026 – Mar 2026

- Designed and evaluated agentic systems using **Gemini** and **Qwen APIs** on TAU-Bench, analyzing **planning depth, tool calling**, and multi-step reasoning failures.
- Proposed and tested a **multi-agent architecture** to improve task completion rate over single-agent baseline.

Context-Aware Summarization Agents (ASU)

Feb 2025 – May 2025

- Built **multi-agent** summarization pipeline using **OpenAI and Gemini APIs** for financial QA (TATQA), improving accuracy by **25%** through **orchestration logic** and prompt design.

Elastic Face Recognition Pipeline (ASU)

Mar 2025 – May 2025

- Deployed **ML inference service** using **FastAPI, PyTorch, AWS EC2** autoscaling, **S3**, and **SQS** task queues.
- Served **100+ concurrent users** with **sub-900ms latency** under load via **AWS autoscaling**.

Technical Skills

Languages: Python, JavaScript, TypeScript, C++, SQL

AI & ML: LLMs (OpenAI, Gemini, Claude), Agentic Systems, Prompt Engineering, PyTorch, Scikit-learn

Backend & APIs: FastAPI, Flask, Node.js, REST APIs, PostgreSQL, MongoDB, Kafka

Cloud & DevOps: AWS (EC2, S3, SQS, Lambda), Docker, Git, CI/CD

Data: NumPy, Pandas, PySpark, Matplotlib, Jupyter